Quantifying and Communicating Uncertainty in SAR-Based Flood Mapping via Density-Aware Neural Networks and Conformal Risk Control

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***Abstract*—Accurate flood mapping from synthetic aperture radar (SAR) imagery is essential for disaster response and risk management. Although deep learning (DL) models have advanced SAR-based flood mapping, they typically lack mechanisms to quantify and communicate predictive uncertainty, limiting their reliability in high-stakes applications. This study proposes a unified framework that captures and conveys two key sources of uncertainty in flood mapping: uncertainty arising from the model's limited knowledge, and uncertainty caused by noise and ambiguity inherent in the data. The former is estimated via feature density in the latent space of a density-aware neural network, while the latter is quantified using *softmax* entropy. These uncertainties are then communicated through conformal risk control under a user-defined risk level (α, δ): inputs associated with high model-knowledge uncertainty are flagged as out-of-distribution (OOD) and abstained from prediction, while data-related uncertainty guides the construction of set-valued predictions for in-distribution (ID) samples. Experiments on diverse real-world flood scenarios—including flooded built-up (FB) areas, flooded vegetation (FV), and flooded bare soil—demonstrate substantial gains in uncertainty quantification and predictive reliability. The proposed density-aware neural network achieves an average area under the receiver operating characteristic (AUROC) of 0.921 in OOD detection, outperforming Bayesian neural networks (BNNs) (0.684) and deep ensembles (DEs) (0.603). OOD abstention improves predictive safety, reducing the average water false negative rate from 36.9% to 28.5% under a risk level of ($\alpha = 0.05$ and $\delta = 0.1$), while prediction sets reduce average miscoverage from 4.4% to 1.6% compared to standard singleton predictions. By explicitly quantifying and communicating the uncertainty in DL predictions, this work strengthens trustworthy, risk-informed decision support for SAR-based flood mapping.**

***Index Terms*—Aleatoric uncertainty (AU), conformal risk control, density-aware neural networks, epistemic uncertainty (EU), risk-informed decision-making, synthetic aperture radar (SAR) flood mapping, uncertainty quantification.**

I. Introduction

FLOODING is among the most widespread and destructive natural disasters, posing significant threats to human lives, infrastructure, economies, and ecosystems. Timely and reliable information during flood emergencies is essential for mitigating damage and preventing loss of life and property. Earth Observation (EO) technologies provide critical data for rapid flood extent mapping and continuous monitoring across large geographic areas. Among various EO sensors, the synthetic aperture radar (SAR) imaging system is the most widely used for flood mapping due to its all-weather and illumination-independent imaging capability. A long history exists of utilizing SAR data from a variety of satellite missions for flood mapping and monitoring, including RADARSAT [1], COSMO-SkyMed [2], TerraSAR-X [3], ALOS2/PALSAR-2 [4], and Sentinel-1 [5]. SAR-based flood mapping has evolved from rule-based and statistical methods [6], [7], through traditional machine learning (ML) approaches [8], [9], and more recently, to advanced deep learning (DL) models [10], [11], [12], [13]. Modern DL models have demonstrated superior performance due to their ability to learn complex nonlinear patterns and scale to large datasets [14]. However, in real-world flood response and management, high classification accuracy alone is insufficient; assessing the prediction reliability of the mapping results is also essential to facilitate effective decision-making. Uncertainty persists in flood mapping regardless of the method employed, making it essential to quantify and communicate the uncertainty associated with the detected flood extents. The global flood mapping (GFM) system, a component of the Copernicus Emergency Management Service (EMS), utilizes ensemble likelihood to illustrate the uncertainty in flood mapping [15]. The work by [16] accounted for SAR image uncertainty in flood mapping with the bootstrap method. Subsequent work by the same team characterized uncertainty using probabilistic flood mapping [17]. Similarly, the work by [4] quantified uncertainty in both SAR intensity and InSAR coherence through a Bayesian network framework for flood mapping in urban environments. However, these studies primarily accounted for uncertainty sourced from input data. A recent study employed a Bayesian neural network (BNN) to quantify the uncertainty associated with predicted flood extent while also accounting for model uncertainty [18]. However, uncertainties from different sources were neither disentangled nor separately quantified. Furthermore, displaying uncertainty merely as probability maps may not be interpretable or actionable for end-users and decision-makers.

When DL is adopted for prediction tasks, the total predictive uncertainty is commonly decomposed into *aleatoric*

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uncertainty (AU) and *epistemic uncertainty* (EU). AU refers to an intrinsic notion of randomness and cannot be reduced by additional training data [19], [20]. EU reflects the model's limited knowledge—due to finite training data or model misspecification—and can, in principle, be reduced by collecting more training data or improving the model's hypothesis space [20], [21]. SAR intensity-based open (flood) water mapping relies on the physical principle that smooth water surfaces produce specular reflections, resulting in low backscatter and appearing as dark tones in SAR images. This behavior generally enables open (flood) water to be distinguished from other land cover types: bare soil typically produces diffuse surface scattering, vegetation exhibits volume scattering, and built-up areas often generate double-bounce scattering, leading to brighter image tones. Despite these distinguishing patterns, SAR-based flood mapping presents several well-known practical challenges. The inherent speckle of SAR data introduces random fluctuations in pixel brightness, complicating classification. In addition, surfaces such as paved urban areas and dryland terrain in arid or semi-arid regions can produce low backscatter values similar to open water. Conversely, rough water surfaces disturbed by wind or rainfall can appear unusually bright. These forms of data noise and class-overlap create ambiguity in the mapping from SAR observations to (flood) water labels and therefore contribute to *AU*. In this sense, the relationship between SAR observations and flood state is treated as stochastic from the model's perspective, even though the underlying physical state is deterministic. More challenging scenarios include flooded urban environments or partially submerged vegetation, where "double-bounce" backscattering creates very bright signatures in SAR images. Models trained primarily on open water examples struggle to generalize to such cases. These failures arise from *EU*, which stems from the model's limited knowledge or insufficient exposure to these types of samples.

To achieve robust and trustworthy flood mapping across diverse land cover types, geographic regions, and environmental conditions, it is essential to disentangle, quantify, and appropriately address both AU and EU. However, quantifying uncertainty alone is insufficient. Equally important—yet often overlooked—is the effective communication of uncertainty to end-users and decision-makers [22]. Without actionable and interpretable uncertainty information, even accurate estimates may fail to support risk-informed decisions. To the best of our knowledge, the rigorous separation of uncertainty sources and the conversion of uncertainty into operationally meaningful forms remain open challenges in the current flood mapping research and practice. To address these challenges, we propose a flexible framework for uncertainty quantification and communication in SAR-based flood mapping. The framework consists of two key components as follows.

1) *Density-Aware Neural Network:* This model captures AU via the entropy of softmax outputs and EU via latent feature density in the representation space.
2) *Conformal Risk Control:* This component translates EU into principled abstention on out-of-distribution (OOD) samples and converts AU into calibrated prediction sets under a user-defined risk level.

To evaluate the proposed framework, we conduct experiments on multiple real-world flood events across diverse landscapes, including flooded urban areas, vegetation, and bare soil. Results show that our method substantially improves EU quantification compared to BNNs [23] and deep ensembles (DEs) [24], while providing enhanced predictive reliability through OOD abstention and calibrated prediction sets. These results highlight the framework's potential for trustworthy, risk-informed flood mapping in operational settings. The rest of this article is organized as follows. Section II provides an overview of related work. Section III details the proposed methodology, while Section IV describes the datasets and experimental setup. Results are presented in Section V, followed by a discussion in Section VI. Finally, concluding remarks are summarized in Section VII.

II. Related Work

This section provides a brief overview of prior works relevant to the proposed framework for risk-informed SAR-based flood mapping. We organize the review into three key areas: 1) BNNs for uncertainty quantification, which are widely adopted across various applications; 2) conformal prediction, which offers principled ways to generate calibrated prediction sets under uncertainty; and 3) OOD detection, which is closely related to EU quantification.

A. BNNs for Uncertainty Quantification

BNNs are widely used for uncertainty quantification [25]. They formalize model uncertainty by imposing a prior distribution $p(\theta)$ over the network parameters and inferring the posterior distribution $p(\theta|\mathcal{D})$ given the training data $\mathcal{D}$. However, exact Bayesian inference is intractable for deep neural networks, necessitating the use of approximation techniques in practice. One popular approximation is Monte Carlo (MC) dropout [23], which retains dropout layers during both training and testing. By performing multiple stochastic forward passes with different dropout masks, MC dropout approximates Bayesian model averaging (BMA) and enables uncertainty estimation with minimal implementation overhead. Another widely used approach for uncertainty quantification in deep neural networks is DEs [24], which trains multiple networks with identical architectures but different random initializations. Recent studies have interpreted DEs as approximate Bayesian methods and shown that they can provide strong empirical performance for both predictive accuracy and uncertainty estimation [26], [27], [28].

In BNNs, the AU is typically quantified via the expected entropy of the softmax distribution $\mathbb{E}_{p(\theta|\mathcal{D})}[\mathbb{H}[p(y|x,\theta)]]$, whereas EU is captured by the mutual information (MI) between model parameters θ and output y

$$\begin{aligned}&\mathbb{I}\left[y,\theta|x,\mathcal{D}\right]\\&=\mathbb{H}\left[\mathbb{E}_{p(\theta|\mathcal{D})}\left[p\left(y|x,\theta\right)\right]\right]-\mathbb{E}_{p(\theta|\mathcal{D})}\left[\mathbb{H}\left[p\left(y|x,\theta\right)\right]\right].\end{aligned}\tag{1}$$

In practice, these expectations are approximated by averaging predictions from multiple stochastic forward passes (in MC Dropout) or across ensemble members (in DEs).

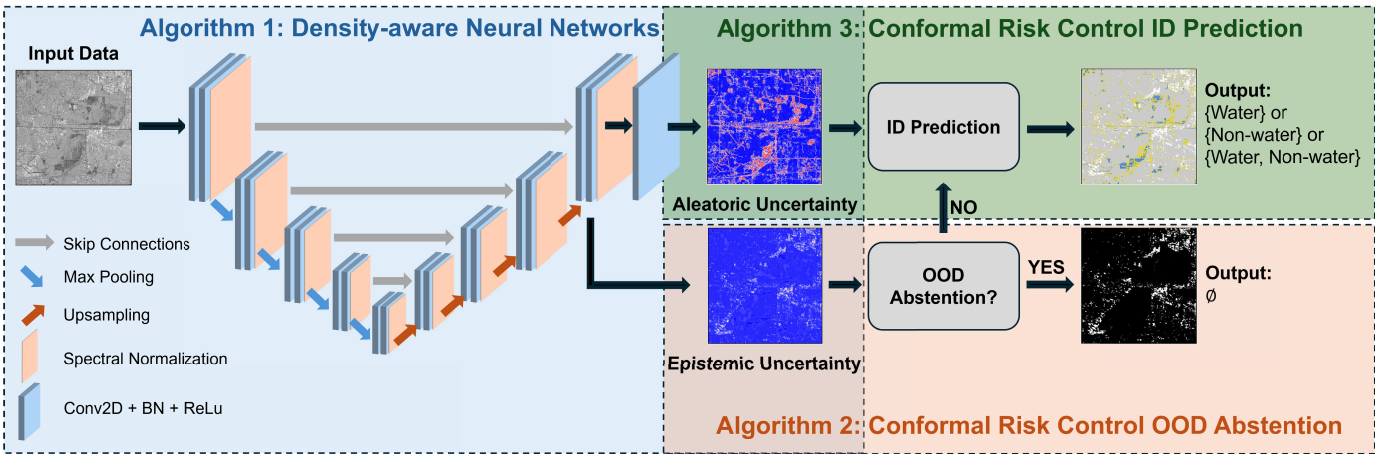

Fig. 1. Workflow of the proposed framework for uncertainty quantification and communication in SAR-based flood mapping.

B. Conformal Prediction

Conformal prediction is a distribution-free inference framework for uncertainty quantification that produces prediction sets rather than singleton predictions. It provides finite-sample coverage guarantees under the assumption of data exchangeability, making no assumptions about the underlying data distribution. Inductive conformal prediction (ICP) [29], also referred to as split conformal prediction [30], has been adopted in classification and regression tasks [31]. Given a pretrained model, a held-out calibration dataset $\mathcal{D}_n = \{(X_1, Y_1), \ldots, (X_n, Y_n)\}$, a conformal score function $s(x, y) \in \mathbb{R}$, and a user-specified error level $\alpha \in (0, 1)$, the prediction set for a test input X_{test} is defined as

$$\mathcal{C}(X_{\text{test}}) = \{y : s(X_{\text{test}}, y) \leq \hat{q}\} \tag{2}$$

where

$$\hat{q} = \text{Quantile}(s_1, \ldots, s_n; \lceil (n+1)(1-\alpha) \rceil / n) \tag{3}$$

$s_1 = s(X_1, Y_1), \ldots, s_n = s(X_n, Y_n)$ are calibration scores. Under the exchangeability assumption for $(X_i, Y_i)_{i=1,\ldots,n}$ and $(X_{\text{test}}, Y_{\text{test}})$, the prediction set $\mathcal{C}(X_{\text{test}})$ satisfies the following marginal coverage guarantee:

$$\mathbb{P}(Y_{\text{test}} \in \mathcal{C}(X_{\text{test}})) \geq 1 - \alpha. \tag{4}$$

ICP is attractive due to its minimal statistical assumptions and broad applicability. The choice of score function s is crucial to obtain valid and informative prediction sets. Although coverage guarantees can be trivially satisfied by producing overly large, uninformative sets (e.g., including the entire output space), such sets lack practical value. To address this, several advanced scoring strategies have been proposed to construct tighter prediction sets while still maintaining the required coverage guarantees [32], [33]. However, these approaches often conflate different sources of uncertainty and typically yield only marginal coverage guarantees, rather than the more desirable conditional guarantees.

C. OOD Detection

DL models typically rely on the assumption that training and test data are independently and identically distributed (i.i.d.). Although they often perform well under this assumption, their performance can degrade significantly when exposed to OOD samples—inputs that deviate from the training data distribution. Such deviations may arise from *semantic shift* (e.g., unseen new semantic classes) or *covariate shift* (e.g., label preserving transformations) [34]. OOD inputs typically exhibit elevated EU, making OOD detection a widely adopted proxy for evaluating a model's EU estimation. OOD detection has been extensively studied across different methodological paradigms, including deep generative models that estimate data likelihoods [35], density-based methods that model latent feature distributions [36], and information-bottleneck frameworks that seek compact representations sensitive to distributional shifts [37]. It is important to emphasize, however, that OOD detection and EU, while related, are conceptually distinct. OOD detection concerns the statistical property of the input data and is often approached using auxiliary detectors in a model-agnostic fashion, potentially independent of the primary task model. In contrast, the EU reflects an inherent property of the model itself—its lack of knowledge or predictive confidence. Therefore, a well-calibrated uncertainty model should assign high EU to OOD inputs that are likely to impair prediction accuracy, while maintaining low uncertainty for in-distribution (ID) inputs. Achieving this alignment between uncertainty estimation and generalization behavior is crucial for trustworthy decision-making in high-stakes applications such as flood mapping.

III. Method

This section introduces the proposed risk-informed flood mapping framework. Section III-A formulates the problem and outlines the types of uncertainty involved. Section III-B presents the density-aware neural network architecture, designed to disentangle and quantify both AU and EU. Section III-C describes how these uncertainties are integrated into a conformal risk control framework, which transforms them into abstentions or prediction sets with formal statistical guarantees. These guarantees are governed by a user-specified risk control level (α, δ), where α denotes the risk tolerance, controlling the maximum allowable fraction of incorrect predictions,

and δ indicates the confidence level, specifying the probability that this risk bound is satisfied. The overall workflow for uncertainty quantification and communication is illustrated in Fig. 1.

A. Problem Statement

Predictive uncertainty in ML and DL models arises from multiple sources, including noise, biases, and incomplete information. It is common practice to categorize uncertainty into two principal types: ***aleatoric (aleatory)*** and ***epistemic (epistemology)***. Although widely used, these terms lack precise formal definitions. The word aleatory derives from the Latin *alea*, meaning "dice roll," and refers to the irreducible randomness inherent in data, such as measurement noise or stochastic variability. In contrast, epistemic originates from the Greek $\epsilon\pi\iota\sigma\tau\eta\mu\eta$ (episteme), meaning "knowledge," and denotes uncertainty due to incomplete information or limited knowledge about the model or data-generating process [38]. Probability theory is commonly employed to represent uncertainty. Under an aleatoric interpretation, probability reflects the long-run frequencies of events, while under an epistemic interpretation, it quantifies a degree of belief given limited knowledge [21], [39]. Consequently, in the ML literature, AU is associated with inherent data noise and is generally considered *irreducible*, while EU captures aspects of the model's ignorance, which may be *reduced* with additional data or improved model capacity [20], [40].

In the supervised learning setting, the objective is to learn a predictive model that estimates targets $y \in \mathcal{Y}$ given inputs $x \in \mathcal{X}$, where samples (x, y) are drawn from an unknown joint distribution P(X,Y). The predictive model is defined as a mapping function $f : \mathcal{X} \to \mathcal{Y}$. Given a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$, the risk (i.e., the expected prediction error) of a predictor f is the expected loss

$$R(f) = \mathbb{E}_{P(X,Y)}\left[\ell(Y, f(X))\right]. \tag{5}$$

Learning is performed within a *hypothesis space* $\mathcal{H} \subset \mathcal{F} = \mathcal{Y}^{\mathcal{X}}$, which specifies the allowable model forms. The goal of learning is to identify a hypothesis $h^* \in \mathcal{H}$ that minimizes the true risk

$$h^* = \underset{h \in \mathcal{H}}{\operatorname{argmin}}\, R(h). \tag{6}$$

Since the true distribution P(X,Y) is unknown in practice, h^* cannot be computed directly. Instead, we minimize the empirical risk over a finite training dataset $\mathcal{D} = \{(x_1, y_1), \ldots, (x_N, y_N)\} \in \mathcal{X} \times \mathcal{Y}$

$$R_{\mathcal{D}}(h) = \frac{1}{N}\sum_{i=1}^{N} \ell(y_i, h(x_i)) \tag{7}$$

$$h_{\mathcal{D}} = \underset{h \in \mathcal{H}}{\operatorname{argmin}}\, R_{\mathcal{D}}(h). \tag{8}$$

Following the taxonomy in [20], we decompose predictive uncertainty into three distinct sources: 1) uncertainty due to inherent randomness; 2) uncertainty due to model misspecification; and 3) uncertainty arising from the finite training dataset. The optimal prediction y^* for an instance x is prescribed by the pointwise Bayes predictor f^*, defined as

$$f^*(x) = \underset{y^* \in \mathcal{Y}}{\operatorname{argmin}}\, \mathbb{E}_{P(Y|X=x)}\left[\ell(Y, y^*)\right]. \tag{9}$$

Due to the inherent nondeterministic dependency between X and Y, the output y is modeled as a conditional probability distribution $p(y|x)$, implying that even perfect knowledge of P(X,Y) does not eliminate uncertainty. This is referred to as *AU*. Restricting the predictor to $\mathcal{H}$ rather than $\mathcal{F}$ yields the optimal predictor h^* within $\mathcal{H}$ that may deviate from f^*, a phenomenon known as *model uncertainty* [20], also termed *model bias* [41] or *distributional uncertainty* [42], reflecting model misspecification due to $f^* \notin \mathcal{H}$. A second source of EU arises from the discrepancy between $h_{\mathcal{D}}$ and h^* due to learning from a finite dataset. This is referred to as *approximation uncertainty* [20] or *estimation uncertainty* [21]. Since empirical risk minimization is performed instead of true risk minimization, different versions of $h_{\mathcal{D}}$ may be induced, and this uncertainty can be expressed by *approximation/estimation variance*. Both *model uncertainty* and *approximation uncertainty* characterize the epistemic state of the predictor and are collectively categorized as *EU*. While AU is irreducible, EU may be reduced through improved hypothesis spaces or larger datasets. To establish a risk-aware SAR-based flood mapping system, our goal is to (1) estimate both AU and EU at the pixel level, and (2) communicate these uncertainties in a statistically calibrated and operationally actionable manner, thereby enabling risk-informed decision-making.

B. Density-Aware Neural Network

According to the definitions introduced earlier, AU represents the model's "known-unknown," whereas EU indicates "unknown-unknown"—a type that is inherently more challenging to quantify. Under the i.i.d. assumption in ML, OOD data that exhibit substantial shifts in distribution are theoretically associated with elevated EU. This insight motivates approaches that quantify EU based on the distance between test and training samples, such as SNGP [43] and DUQ [44]. These models leverage Gaussian processes (GPs) or radial basis functions (RBFs) in the output layer, allowing distance-aware predictions with uncertainty estimates via a single forward pass. While effective in image classification benchmarks, these approaches typically require modifications to network architecture and introduce additional hyperparameters. Furthermore, neural networks trained under empirical risk minimization are susceptible to *shortcut learning*, where they exploit spurious yet highly predictive features while discarding task-relevant but less correlated features [45], [46]. This phenomenon can cause OOD inputs to be incorrectly mapped into ID regions of the latent space, leading to overconfident but erroneous predictions—a behavior known as "feature collapse" [47].

To address this, we adopt a strategy inspired by [48] that injects distance-preserving (e.g., by fostering feature density awareness) constraints and inductive bias into image segmentation neural networks (e.g., U-Net) without imposing major architectural changes or additional hyperparameters.

Specifically, we encourage the neural feature extractor f_θ, parameterized by θ, to preserve distances via a bi-Lipschitz constraint, ensuring for all input pairs x_1 and x_2

$$K_1 \|x_1 - x_2\|_I \leq \|f_\theta(x_1) - f_\theta(x_2)\|_F \leq K_2 \|x_1 - x_2\|_I \quad (10)$$

where $\|\cdot\|_I$ and $\|\cdot\|_F$ denote distance metrics in the input and feature spaces, respectively, and K_1 and K_2 are lower and upper Lipschitz constants [43]. This constraint ensures that f_θ satisfies both *sensitivity* (through the lower Lipschitz bound) and *smoothness* (through the upper Lipschitz bound). The lower bound prevents semantically distinct inputs from collapsing into similar features, while the upper bound guards against small, irrelevant perturbations causing drastic changes in feature space.

Algorithm 1 Uncertainty Quantification With Density-Aware Neural Network

Input: Pre-trained density-aware network f_θ, training dataset $(x_i, y_i)_{i=1}^N$, test data x_{test}

Output: Aleatoric Uncertainty (AU), Epistemic Uncertainty (EU)

1: $z_{i=1:N} \leftarrow PCA \leftarrow f_\theta(x_{i=1:N})$
2: $GDA_{\text{train}} \leftarrow GDA(z_{i=1:N}, y_{i=1:N})$
3: $z_{\text{test}} \leftarrow PCA \leftarrow f_\theta(x_{\text{test}})$
4: $EU : q(z_{\text{test}}) \leftarrow GDA_{\text{train}}(z_{\text{test}})$
5: $AU : \mathbb{H}[p(y_{\text{test}}|x_{\text{test}}, \theta)] \leftarrow softmax(f_\theta(x_{\text{test}}))$

The upper bound is directly linked to the Lipschitz constant of each layer. For a neural network composed of weight matrices W_i followed by 1-Lipschitz nonlinearities (e.g., ReLU), the global Lipschitz constant is upper bounded by the product of their spectral norms, i.e., $\text{Lip}(f_\theta) \leq \prod_{i=1} \|W_i\|_2$ [49]. Spectral normalization constrains these norms by replacing each weight matrix W with $W/\sigma(W)$, where $\sigma(W)$ is the spectral norm estimated using power iteration [50]. Compared to other methods to achieve upper Lipschitz bound like gradient penalty [51], spectral normalization provides a favorable tradeoff between computational efficiency and stability for convolutional networks [43], [48]. We combine spectral normalization and residual blocks, which are widely used in modern vision architectures, to enforce the bi-Lipschitz constraint [48]. Specifically, we employ the U-Net [52] image segmentation architecture with a ResNet34 [53] encoder, applying spectral normalization to all convolutional layers except the output layer. This minor adjustment preserves model differentiability and enables standard training via stochastic gradient descent. During inference, AU is quantified via the entropy $\mathbb{H}[Y|x, \theta]$ of the softmax output $p(y|x, \theta)$, whereas EU is quantified using feature density. Since density estimation degrades with increasing dimensionality [54], we perform principal component analysis (PCA) to reduce feature dimensionality before density estimation. Although the feature dimensionality (e.g., 16) is moderate in our case, PCA empirically improves both the robustness and accuracy of the EU estimation. To compute feature density, we employ Gaussian discriminant analysis (GDA), which models the training features as a Gaussian mixture—one Gaussian per class. Let $z = f_\theta(x)$ be the PCA-reduced feature representation of an input x. The feature density $q(z)$ is computed by marginalizing over class-conditional densities, $q(z) = \sum_y q(z|y)q(y)$, where $q(z|y)$ is estimated from GDA and $q(y)$ is the empirical class prior from the training dataset. Inputs mapped to low-density regions are interpreted as having high EU. A schematic of uncertainty quantification using the density-aware U-Net is provided in Fig. 1, and the main steps are summarized in Algorithm 1.

C. Conformal Risk Control

As illustrated in Fig. 1, uncertainty communication under the conformal risk control framework follows a two-stage decision process. For each test sample x, the procedure first performs OOD abstention using feature density $q(z)$ (capturing EU) to determine whether the prediction should be withheld. If the sample is retained, a set-valued prediction is then made based on the softmax output $p(y|x)$ (reflecting AU) to generate the final prediction set.

Risk Control OOD Abstention: Test samples with high EU (i.e., *unknown-unknowns*) are treated as OOD and abstained from prediction to avoid unreliable outcomes. Only ID samples (i.e., *known-unknowns*) should be retained for prediction. This prediction approach follows the selective classification principle [55]. We adopt the conformal inference framework to transfer high EU into abstention decisions. We define the score function as the feature density $s(x) = q(z)$. A lower score $s(x)$ indicates lower density and hence higher EU, making x more likely to be OOD. Given a calibration dataset $\mathcal{Z}_{\text{cal}}$ of n i.i.d. samples $x \sim D_X^n$, the conformal p-value for each test sample can be calculated as

$$\mathrm{p}(x) = \frac{1 + |\{i \in \mathcal{Z}_{\text{cal}} : s(X_i) \leq s(x)\}|}{n+1}. \quad (11)$$

These marginal p-values $\mathrm{p}(X_{\text{test}})$ satisfy the standard marginal validity condition

$$\mathbb{P}(\mathrm{p}(X_{\text{test}}) \leq \alpha) \leq \alpha, \ \alpha \in (0, 1) \quad (12)$$

whenever $X_{\text{test}} \sim D_X$. This is equivalent to an expected guarantee conditional on the calibration data $\mathcal{Z}_{\text{cal}}$: $\mathbb{E}[\mathbb{P}(\mathrm{p}(X_{\text{test}}) \leq \alpha|\mathcal{Z}_{\text{cal}})] \leq \alpha$. In practice, instead of the expected error, one often desires a stronger calibration-conditional guarantee that holds with high probability $1-\delta$ over the randomness of $\mathcal{Z}_{\text{cal}}$

$$\mathbb{P}_{\mathcal{Z}_{\text{cal}} \sim D_X^n}\left(\mathbb{P}\left(\mathrm{p}'(X_{\text{test}}) \leq \alpha\right) \leq \alpha\right) \geq 1 - \delta \quad (13)$$

for any $\delta \in (0, 1)$.

This calibration-conditional guarantee is critical in operational deployments. Suppose multiple practitioners perform OOD abstention using independently drawn calibration datasets. Equation (12) guarantees that, on average over calibration datasets, the false discovery rate (FDR) (i.e., the proportion of ID samples incorrectly flagged as OOD) does not exceed α. Equation (13) further strengthens this result by ensuring that this risk bound holds with probability at least $1 - \delta$ over the randomness of the calibration data. In operational terms, this means that at least a $1-\delta$ proportion of practitioners—each using their own calibration set—can trust that the FDR of the system is bounded by α, providing a precise measure of confidence. This defines the desired (α, δ) risk

control. In large-scale SAR-based flood mapping, predictions are made per pixel, introducing a multiple-hypothesis testing setting in which the marginal p-values calculated from (11) are not statistically independent. As shown in [56] and [57], dependence can lead to anti-conservative behavior in a global null hypothesis test. To maintain the guarantee in (13) under multiple testing, the p-values must be adjusted so that the adjusted values remain calibration-conditional valid. Following [57, Th. 4], a valid adjustment can be achieved through a simultaneous upper confidence bound:

Let $U_1, \ldots, U_n \overset{\text{i.i.d.}}{\sim} \text{Unif}([0,1])$, with order statistics $U_{(1)} \leq U_{(2)} \leq \cdots \leq U_{(n)}$, and $\delta \in (0,1)$. Suppose $0 \leq b_1 \leq b_2 \leq \cdots \leq b_n \leq 1$ are n reals such that

$$\mathbb{P}\left(U_{(1)} \leq b_1, \ldots, U_{(n)} \leq b_n\right) \geq 1 - \delta. \tag{14}$$

Define $b_0 = 0, b_{n+1} = 1$, and a piecewise constant function $h : [0,1] \to [0,1]$ as

$$h(t) = b_{\lceil (n+1)t \rceil}, \quad t \in [0,1]. \tag{15}$$

Then, the adjusted p-values $\mathrm{p}' = h(\mathrm{p})$ satisfy (13), i.e., $\mathrm{p}'(X_{\text{test}})$ is a calibration-conditional valid p-value.

Multiple choices of adjustment exist. In SAR-based flood mapping, where a large amount of the calibration data n is typically available, we apply the asymptotic adjustment method [57]

$$b_i = \min\left\{\frac{i}{n} + c_n(\delta)\frac{\sqrt{i(n-i)}}{n\sqrt{n}}, 1\right\}, \quad i = 1, \ldots, n \tag{16}$$

where

$$c_n(\delta) = \frac{-\log\left[-\log(1-\delta)\right] + 2\log\log n + \frac{1}{2}\log\log\log n}{\sqrt{2\log\log n}} - \frac{\frac{1}{2}\log\pi}{\sqrt{2\log\log n}}.$$

Algorithm 2 Conformal Risk Control OOD Abstention

Input: Pre-trained density-aware network f_θ, calibration dataset $\mathcal{Z}_{\text{cal}}$, risk control level (α, δ), test data x_t

Output: OOD indicator $OOD_t \in \{0, 1\}$

```
for t ← 1, 2, ... do
    p_t ← Eq.(11)
    p'_t ← Eqs.(15), (16)
    if p'_t ≤ α then
        OOD_t ← 1 ▷Flag as OOD
    else
        OOD_t ← 0 ▷Retain for prediction
    end if
end for
```

The complete procedure for performing risk control OOD abstention with calibration-conditional valid p-value is outlined in Algorithm 2.

Risk Control ID Prediction: Pixels in the test image that are not flagged as OOD are retained for prediction under the conformal risk control. Given a predefined risk level and the model's softmax output (used to estimate AU), each pixel is assigned a set-valued prediction: {Water}, {Nonwater}, or {Water, Nonwater}. Let $\mathcal{C}_\lambda$ denote a set-valued predictor parameterized by λ, such that $\mathcal{C}_\lambda(X) \subseteq \mathcal{Y}$ specifies the prediction set for the target variable Y. The size of the prediction set is determined by λ. Given a loss function $L(y, \mathcal{C}_\lambda(x))$, the expected risk is defined as

$$R(\mathcal{C}_\lambda) = \mathbb{E}\left[L\left(Y_{\text{test}}, \mathcal{C}_\lambda(X_{\text{test}})\right)\right]. \tag{17}$$

Suppose we have a calibration dataset $\mathcal{S}_{\text{cal}}$ consisting of n i.i.d. samples $(x, y) \sim D^n$, which yields a confidence set predictor $\mathcal{C}_{\hat{\lambda}}$. We say that $\mathcal{C}_{\hat{\lambda}}$ is *(α, δ) risk-controlling*, or satisfies the *probably approximately correct (PAC)* guarantee [58], if

$$\mathbb{P}_{\mathcal{S}_{\text{cal}} \sim D^n}\left(R\left(\mathcal{C}_{\hat{\lambda}}\right) \leq \alpha\right) \geq 1 - \delta. \tag{18}$$

The objective is to find $\hat{\lambda}$ that yields the smallest possible prediction set (i.e., the most informative) that satisfies (18). In pixel-wise image segmentation for flood mapping, we adopt the miscoverage risk

$$L\left(y, \mathcal{C}_\lambda(x)\right) = \mathbb{1}\left\{y \notin \mathcal{C}_\lambda(x)\right\}. \tag{19}$$

The prediction set $\mathcal{C}_\lambda$ is constructed by thresholding the softmax output

$$\mathcal{C}_\lambda(x) = \left\{y : p(y|x, \theta) \geq 1 - \lambda\right\}. \tag{20}$$

This creates nested confidence sets—smaller λ produces smaller sets and greater miscoverage risk

$$\lambda \leq \lambda' \Rightarrow \forall x, \mathcal{C}_\lambda(x) \subset \mathcal{C}_{\lambda'}(x) \Rightarrow L\left(y, \mathcal{C}_\lambda(x)\right) \geq L\left(y, \mathcal{C}_{\lambda'}(x)\right). \tag{21}$$

The empirical risk on the calibration set is computed as

$$\hat{R}\left(\mathcal{C}_\lambda\right) = \frac{1}{n}\sum_{(x,y)\in\mathcal{S}_{\text{cal}}} \mathbb{1}\left\{y \notin \mathcal{C}_\lambda(x)\right\}. \tag{22}$$

A valid pointwise upper confidence bound for the true risk $R(\mathcal{C}_\lambda)$ for each λ, based on the empirical risk and calibration dataset, satisfies

$$\mathbb{P}_{\mathcal{S}_{\text{cal}} \sim D^n}\left(R\left(\mathcal{C}_\lambda\right) \leq \hat{R}^+\left(\mathcal{C}_\lambda\right)\right) \geq 1 - \delta. \tag{23}$$

The optimal parameter $\hat{\lambda}$ is then

$$\hat{\lambda} = \inf\left\{\lambda : \hat{R}^+\left(\mathcal{C}_\lambda\right) \leq \alpha\right\}. \tag{24}$$

A simple but conservative choice of $\hat{R}^+(\mathcal{C}_\lambda)$ is the Vapnik–Chervonenkis (VC) dimension-based generalization bound [59], [60]

$$\mathbb{P}_{\mathcal{S}_{\text{cal}} \sim D^n}\left(R\left(\mathcal{C}_\lambda\right) \leq \underbrace{\hat{R}\left(\mathcal{C}_\lambda\right) + \sqrt{\frac{\log(2n) + 1 - \log(\delta/4)}{n}}}_{\hat{R}^+_{\text{VC}}(\mathcal{C}_\lambda)}\right) \geq 1 - \delta. \tag{25}$$

Given the Bernoulli nature of the loss (0 or 1), a tighter binomial-based bound can be derived using the Bentkus inequality [61], [62]

$$\mathbb{P}\left(\hat{R}\left(\mathcal{C}_\lambda\right) \leq \alpha\right) = \mathbb{P}\left(\text{Binom}\left(n, R\left(\mathcal{C}_\lambda\right)\right) \leq \lceil n\alpha \rceil\right). \tag{26}$$

Thus, a tighter and more accurate risk upper bound is

$$\hat{R}^+_{\text{Bin}}\left(\mathcal{C}_\lambda\right) = \sup\left\{R : \mathbb{P}\left(\text{Binom}(n, R) \leq \left\lceil n\hat{R}\left(\mathcal{C}_\lambda\right)\right\rceil\right) \geq \delta\right\}. \tag{27}$$

Algorithm 3 Conformal Risk Control ID Prediction

Input: Pre-trained density-aware network f_θ, calibration dataset $\mathcal{S}_{\text{cal}}$, risk control level (α, δ), in-distribution test data x_t

Output: Prediction set $\mathcal{C}_\lambda^t$

$\hat{R}^+(\mathcal{C}_\lambda) \leftarrow \emptyset$
$\lambda \leftarrow 1$
while $\lambda > \zeta$ **do**
  $\lambda \leftarrow \lambda - \zeta$ ▷ ζ is the step size for $\hat{\lambda}$ searching
  $\hat{R}(\mathcal{C}_\lambda) \leftarrow$ Eqs.(20), (22)
  $\hat{R}^+_{\text{Bin}}(\mathcal{C}_\lambda) \leftarrow$ Eq. (27)
  $\hat{R}^+(\mathcal{C}_\lambda) \leftarrow \hat{R}^+(\mathcal{C}_\lambda) \cup \hat{R}^+_{\text{Bin}}(\mathcal{C}_\lambda)$
end while
$\hat{\lambda} \leftarrow Eq.(24)$
for $t \leftarrow 1, 2, \ldots$ **do**
  $\mathcal{C}_\lambda^t \leftarrow \emptyset$
  for all $y \in \mathcal{Y}$ **do**
    if $p(y|x_t, \theta) \geq 1 - \hat{\lambda}$ **then**
      $\mathcal{C}_\lambda^t \leftarrow \mathcal{C}_\lambda^t \cup y$
    end if
  end for
end for

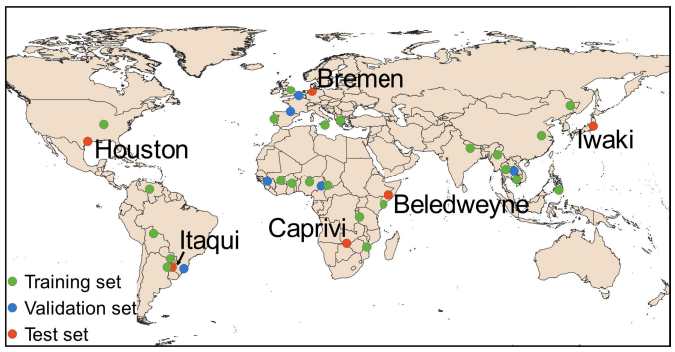

Fig. 2. Geographical distribution of flood events used for model training, calibration, and testing.

The final value of $\hat{\lambda}$ is obtained by iteratively evaluating this bound. The entire procedure of risk control ID prediction is summarized in Algorithm 3.

IV. Datasets and Experiments

This study employs Sentinel-1 ground range detected (GRD) data. The geographical distribution of the training, validation, and test datasets across various flood events is illustrated in Fig. 2. Water reference masks—including both permanent water bodies and floodwater—for the training datasets were initially sourced from the Copernicus GFM service. GFM provides global flood and surface water monitoring using Sentinel-1 VV-polarized imagery, based on an ensemble of three different algorithms and outputs binary masks of open water [15]. To ensure label quality, a visual inspection step was conducted to remove low-quality masks, retaining only high-confidence water references (i.e., those without substantial overestimation or underestimation based on visual assessment). Reference data for the validation and test sets were manually annotated using cloud-free Sentinel-2 optical data acquired as close as possible to the corresponding Sentinel-1 acquisitions. All SAR data used in this study are preprocessed VV-polarized Sentinel-1 GRD products obtained through Google Earth Engine (GEE). Both the SAR imagery and corresponding water reference masks have a spatial resolution of 20 m. The DL model is designed to provide functionality analogous to the GFM service by producing binary open surface water segmentation. The training and validation datasets primarily consist of flood events dominated by open floodwater over bare soil. In contrast, the six test flood events encompass a broader range of flood conditions: four events—the Houston flood (USA, August 30, 2017), the Iwaki flood (Japan, October 12, 2019), the Itaqui flood (Brazil, November 23, 2023), and the Beledweyne flood (Somalia, October 30, 2019)—contain both flooded built-up (FB) areas and flooded bare soils. The Caprivi flood (Namibia, April 06, 2017) features flooded vegetation (FV) alongside flooded bare soils. The Bremen flood (Germany, January 06, 2024) involves only flooded bare soils.

For the flood mapping model, we adopt the U-Net architecture with a ResNet34 encoder (UNet34), which has demonstrated strong performance in SAR-based flood and surface water segmentation tasks [14], [63]. The model was trained using the Adam optimizer [64] with an initial learning rate of 10^{-4} and a weighted cross-entropy loss. Training was performed over 200 epochs with a batch size of 16, and the learning rate was decayed using cosine annealing [65]. To evaluate uncertainty quantification, MC dropout [23] and DEs [24] were compared using the same UNet34 model without spectral normalization. MC dropout was implemented through 20 stochastic forward passes at inference, whereas DEs consisted of ten independently trained models. For the EU estimation with the density-aware neural network, we applied PCA with three principal components. For OOD abstention experiments, a calibration dataset $\mathcal{Z}_{\text{cal}}$ was created by randomly sampling 10% of the validation data (approximately 2 million pixels). To calibrate the model for ID predictions and mitigate class imbalance—particularly the predominance of nonwater pixels—we constructed a class-balanced calibration subset $\mathcal{S}_{\text{cal}}$ by sampling 100 000 pixels per class (water and nonwater) from the validation dataset. All models were implemented in Pytorch, and all experiments were carried out on a workstation equipped with an Intel Core i9-9820X CPU and an NVIDIA RTX A6000 GPU.

V. Results

A. Performance of Uncertainty Quantification

Since the model was trained on single-date SAR images, it was designed to detect surface water in general, covering both permanent water bodies and floodwater. The performance of the density-aware network, MC Dropout, and DEs in water detection was evaluated using overall accuracy (OA), intersection over union (IoU), precision, recall, and $F1$-score. Two evaluation strategies were employed: 1) *overall water*, which includes all surface water types (open water, FB areas, and FV); and 2) *open water*, which excludes FB and FV pixels to isolate performance on bare soil floodwater. AU quantification

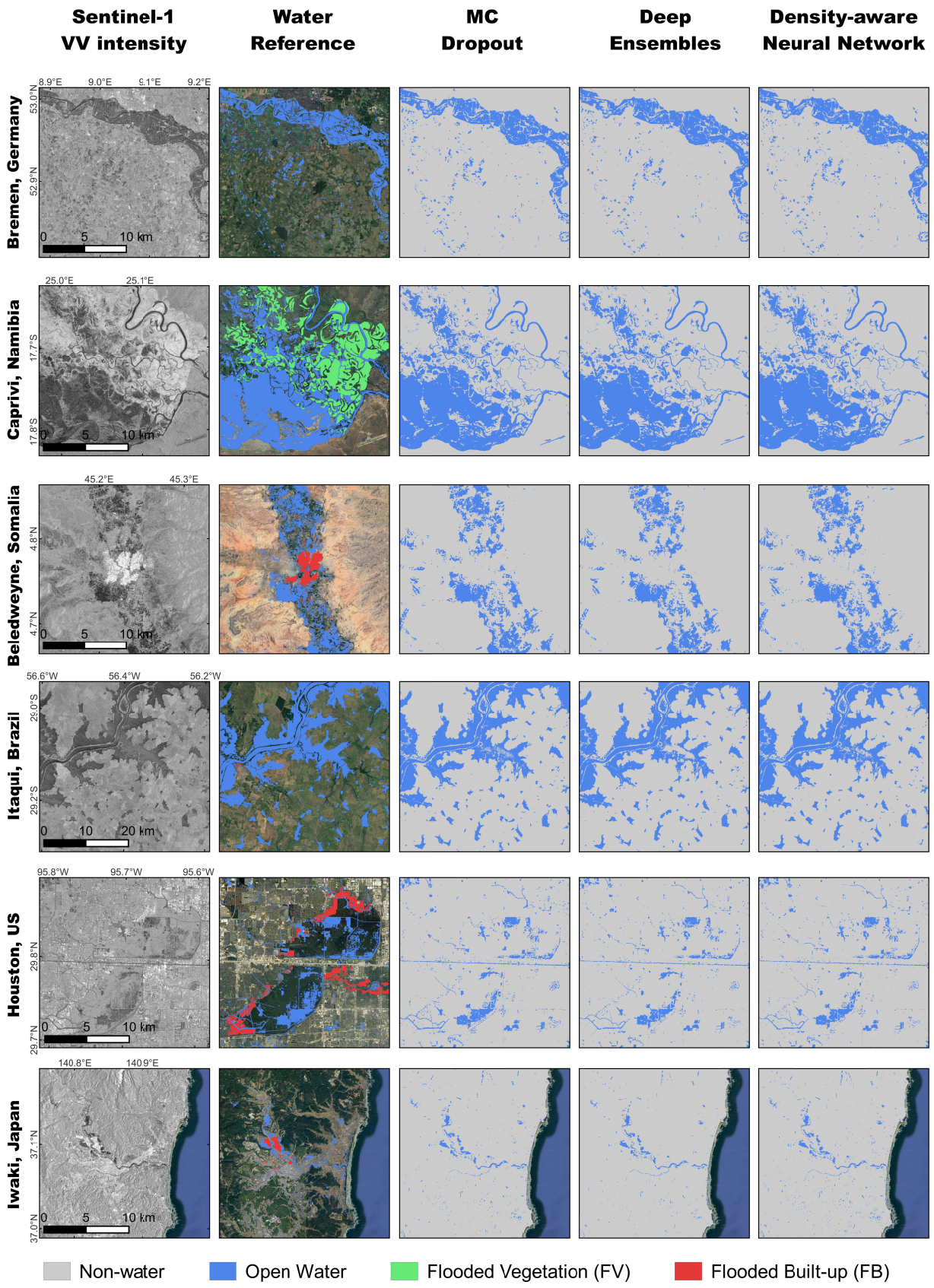

Fig. 3. Water detection results produced by different models. All models misclassified FB and FV as nonwater.

was assessed using the expected calibration error (ECE), while EU was evaluated via the area under the receiver operating characteristic (AUROC) curve, which measures the model's ability to distinguish ID from OOD samples. As no explicit

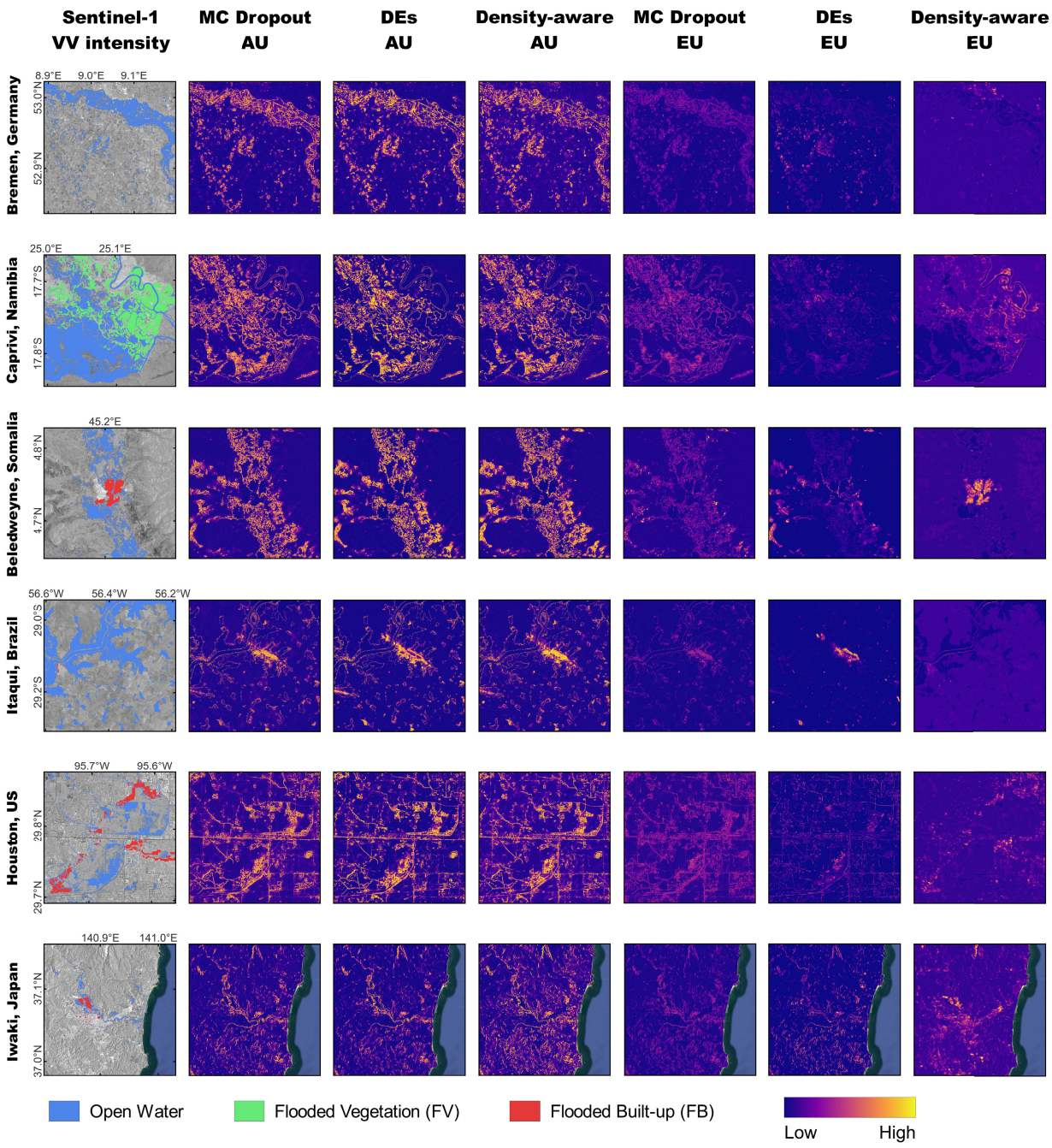

Fig. 4. Uncertainty quantification results from different models. AU exhibits similar spatial patterns across all methods, whereas EU varies substantially. Only the proposed density-aware neural network successfully identifies elevated EU in FB and FV regions.

ground truth is available for distinguishing ID from OOD samples, we adopt a task-dependent definition of OOD that is relative to the training distribution rather than the physical occurrence of flooding. The model is trained as an open water detector, where open floodwater is characterized by specular SAR backscatter. In contrast, FB and FV areas exhibit distinct scattering mechanisms (e.g., predominantly strong double-bounce, although not universally) and are treated as separate semantic categories in SAR-based flood mapping [66], [67]. As a result, FB and FV pixels introduce novel categories and a distributional shift with respect to the training data and modeling objective, and are therefore regarded as task-OOD samples for the trained model. Under this formulation, FB and FV regions are expected to induce elevated EU. Accordingly, the AUROC for separating FB/FV pixels from the remainder of the scene is used as an indicator of the quality of EU estimation. Table I summarizes the quantitative evaluation of the results, and Figs. 3 and 4 show the corresponding water detection and uncertainty maps. The results indicate comparable performance in water detection across all models. DEs and MC Dropout slightly outperform the density-aware model in terms of OA, IoU, and $F1$-score: DEs achieve an average OA

TABLE I

QUANTITATIVE EVALUATION OF (FLOOD) WATER DETECTION AND UNCERTAINTY QUANTIFICATION ACROSS DIFFERENT MODELS

Test Site	Model	Overall Water					Open Water					ECE (↓)	AUROC (↑)
		OA(%)	IoU	Recall	Precision	F1	OA(%)	IoU	Recall	Precision	F1		
Bremen, Germany	MC Dropout	-	-	-	-	-	96.6	0.764	0.870	0.863	0.866	0.007	-
	Deep Ensembles	-	-	-	-	-	**96.7**	**0.773**	0.871	**0.873**	**0.872**	**0.004**	-
	Density-aware Neural Network	-	-	-	-	-	**96.7**	**0.773**	**0.886**	0.858	**0.872**	0.006	-
Caprivi, Namibia	MC Dropout	76.7	**0.565**	**0.596**	0.916	**0.722**	92.4	**0.827**	**0.895**	0.916	**0.905**	**0.016**	0.770
	Deep Ensembles	**76.8**	0.561	0.585	**0.934**	0.719	**92.5**	0.826	0.878	**0.934**	**0.905**	0.020	0.644
	Density-aware Neural Network	76.7	0.562	0.589	0.925	0.720	92.4	0.825	0.885	0.925	0.904	0.027	**0.830**
Beledweyne, Somalia	MC Dropout	93.0	0.543	0.755	0.659	0.704	94.8	0.619	0.911	0.659	0.765	0.011	0.759
	Deep Ensembles	**93.5**	**0.563**	0.752	**0.692**	**0.721**	**95.3**	**0.646**	0.907	**0.692**	**0.785**	**0.007**	0.655
	Density-aware Neural Network	92.7	0.537	**0.761**	0.646	0.699	94.5	0.611	**0.918**	0.646	0.758	0.014	**0.990**
Itaqui, Brazil	MC Dropout	**98.3**	**0.930**	**0.964**	0.963	**0.964**	**98.4**	**0.933**	**0.967**	0.964	**0.975**	0.008	0.593
	Deep Ensembles	97.9	0.914	0.942	**0.969**	0.955	98.0	0.917	0.944	**0.969**	0.957	**0.003**	0.319
	Density-aware Neural Network	**98.3**	**0.930**	**0.964**	0.964	**0.964**	**98.4**	**0.933**	**0.967**	0.964	0.965	0.007	**0.975**
Houston, US	MC Dropout	**89.6**	**0.336**	**0.378**	0.759	**0.483**	**93.8**	**0.471**	**0.561**	0.747	**0.641**	**0.012**	0.644
	Deep Ensembles	89.3	0.306	0.338	**0.763**	0.469	93.6	0.434	0.502	**0.763**	0.605	0.016	0.681
	Density-aware Neural Network	89.5	0.318	0.354	0.759	**0.483**	93.7	0.450	0.525	0.759	0.621	0.017	**0.890**
Iwaki, Japan	MC Dropout	97.5	0.358	0.473	0.596	0.527	98.3	0.441	0.631	0.594	0.612	0.011	0.709
	Deep Ensembles	**97.8**	**0.368**	0.448	**0.675**	**0.538**	**98.5**	**0.462**	0.596	**0.672**	**0.632**	**0.010**	0.714
	Density-aware Neural Network	97.3	0.346	**0.488**	0.544	0.514	98.0	0.420	**0.651**	0.542	0.591	0.015	**0.922**
Average	MC Dropout	91.9	**0.583**	0.673	0.791	**0.714**	95.7	**0.676**	**0.806**	0.790	0.792	0.011	0.684
	Deep Ensembles	**92.0**	0.581	0.656	**0.818**	0.712	**95.8**	**0.676**	0.783	**0.817**	**0.793**	**0.010**	0.603
	Density-aware Neural Network	91.9	0.578	**0.674**	0.783	0.709	95.6	0.669	0.805	0.782	0.785	0.014	**0.921**

TABLE II

EVALUATION OF OOD ABSTENTION PERFORMANCE ACROSS VARYING α VALUES WITH FIXED $\delta = 0.1$

Study Site	Risk Tolerance Level	OOD Abstention Power	OOD FDR	Water FNR	Water FNR (wo OOD Abstention)
Caprivi, Namibia	$\alpha = 0.1$	48.8%	4.5%	29.8%	41.1%
	$\alpha = 0.05$	33.7%	2.1%	33.8%	
	$\alpha = 0.01$	5.4%	0.4%	40.1%	
Beledweyne, Somalia	$\alpha = 0.1$	96.5%	5.4%	8.7%	23.9%
	$\alpha = 0.05$	92.0%	2.2%	9.7%	
	$\alpha = 0.01$	67.1%	0.3%	14.0%	
Itaqui, Brazil	$\alpha = 0.1$	82.5%	2.1%	3.4%	3.6%
	$\alpha = 0.05$	56.4%	0.9%	3.5%	
	$\alpha = 0.01$	8.1%	0.2%	3.6%	
Houston, US	$\alpha = 0.1$	71.1%	4.8%	54.2%	64.6%
	$\alpha = 0.05$	50.7%	1.1%	57.5%	
	$\alpha = 0.01$	12.2%	0.1%	63.1%	
Iwaki, Japan	$\alpha = 0.1$	89.6%	14.7%	37.4%	51.2%
	$\alpha = 0.05$	83.6%	6.9%	38.2%	
	$\alpha = 0.01$	53.8%	0.9%	43.4%	
Average	$\alpha = 0.1$	77.7%	6.3%	26.7%	36.9%
	$\alpha = 0.05$	63.3%	2.6%	28.5%	
	$\alpha = 0.01$	29.3%	0.4%	32.9%	

of 92.0% (95.8%), IoU of 0.581 (0.676), and $F1$-score of 0.712 (0.793); MC Dropout obtains 91.9% (95.7%), 0.583 (0.676), and 0.714 (0.792); while the density-aware model achieves 91.9% (95.6%), 0.578 (0.669), and 0.709 (0.785), for overall water (open water) detection across the six test events. As expected, all models consistently misclassified FB and FV regions as nonwater, due to their binary training objective of distinguishing only open water from land. Because these flooded land-cover types exhibit backscatter characteristics more similar to land than to open water, the models lacked the representational capacity to identify them correctly. This limitation is particularly evident in the Houston and Iwaki flood events, where the large proportion of FB leads to noticeably lower performance under the overall water evaluation compared to other test cases. Even under the open water evaluation, these two events yield substantially lower IoU and $F1$-scores due to their more complex environments, which introduce both overestimation (e.g., paved surfaces within urban areas or mountain shadows) and underestimation (e.g., medium-backscatter open water inside urban areas). In terms of AU, DEs perform best, with an average ECE of 0.010, followed by MC Dropout (0.011) and the density-aware model (0.014). All models produce similar spatial patterns in AU maps (Figs. 4–7), with elevated uncertainty observed in ambiguous areas such as water–land boundaries, dry floodplains (Beledweyne), roads and pavements (Houston), agricultural fields (Bremen), and wind-roughened water (Itaqui). In contrast, EU maps reveal significant differences across methods. Only the density-aware model consistently highlights high uncertainty in FB and FV regions, accurately signaling the model's unfamiliarity with these scenarios. This behavior provides an explicit epistemic warning: *"I do not know how to classify these areas—please interpret with caution."* Fig. 5 shows the details of this behavior for missed FV in Caprivi, while Figs. 6 and 7 illustrate similar effects over the missed FB areas in Houston and Iwaki, respectively. Table I shows that the density-aware model significantly outperforms the other two in EU quantification, achieving an average AUROC of 0.921, compared to 0.684 for MC Dropout and 0.603 for DEs. MC Dropout and DEs fail to provide such warnings, as their

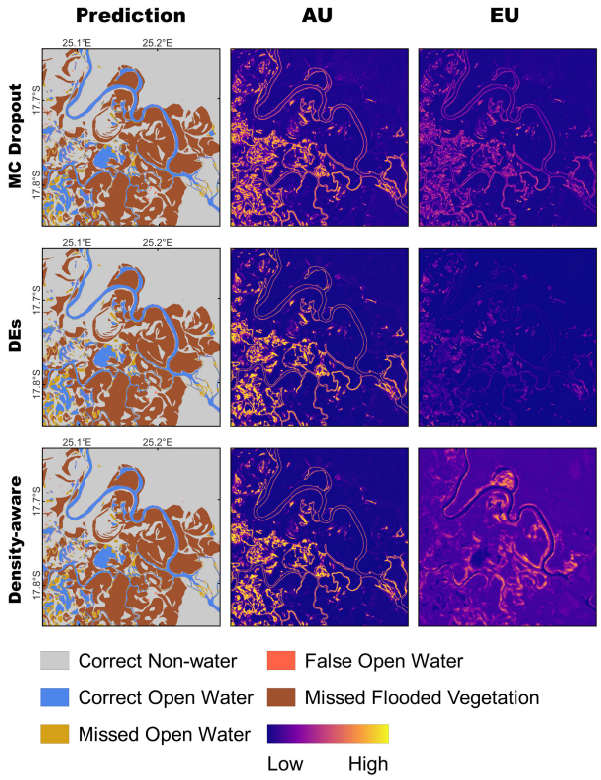

Fig. 5. Comparison of predictions and corresponding uncertainty quantification from different models for a subset of the Caprivi flood event.

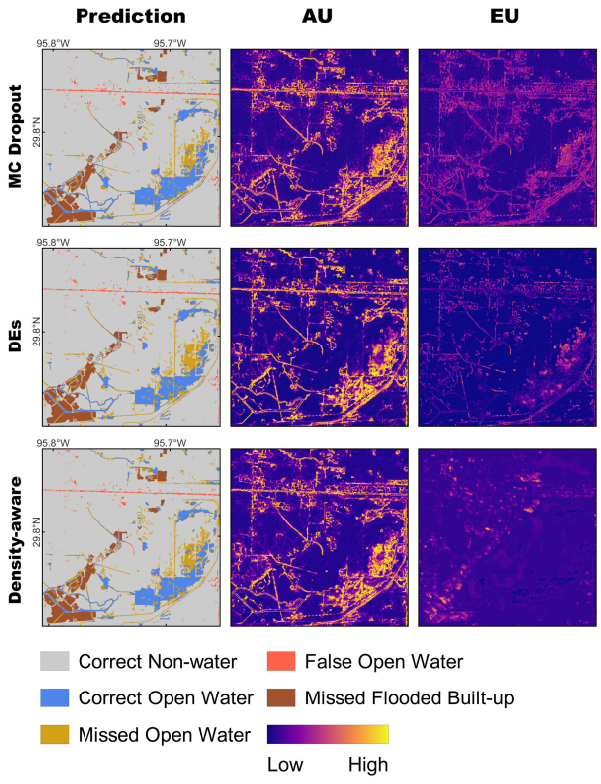

Fig. 6. Comparison of predictions and corresponding uncertainty quantification from different models for a subset of the Houston flood event.

epistemic estimates are derived from variance across stochastic forward passes or ensemble members. In FB and FV regions, all forward passes of MC Dropout and DEs yield consistently

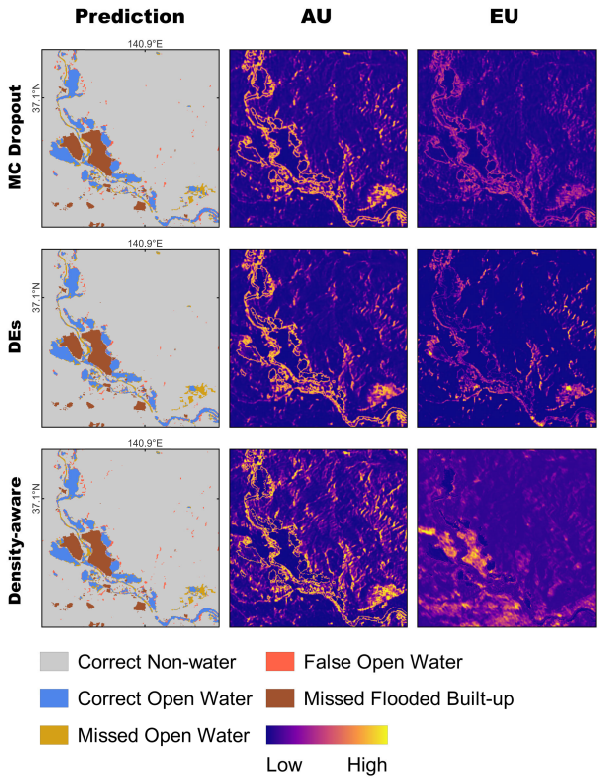

Fig. 7. Comparison of predictions and corresponding uncertainty quantification from different models for a subset of the Iwaki flood event.

incorrect nonwater predictions; the ensemble variance is low, resulting in low EU—despite the model lacking exposure to such cases during training.

B. Predictions Under Conformal Risk Control

Fig. 8 presents the results of OOD abstention and ID prediction sets across six test flood events at a nominal risk control level of ($\alpha = 0.05$ and $\delta = 0.1$). Table II provides a quantitative evaluation of the OOD abstention performance using two key metrics: OOD abstention power and OOD FDR. The OOD abstention power measures the proportion of abstained FB and/or FV samples, indicating how effectively the model avoids incorrect predictions in challenging OOD regions. Conversely, OOD FDR reflects the proportion of abstained open water samples, capturing the negative side effect of abstaining in regions where the model could have produced correct predictions. As shown earlier, the open water detection network tends to misclassify FB and FV areas as nonwater, leading to a high false negative rate (FNR). OOD abstention aims to mitigate this issue by excluding predictions in these regions, thereby reducing overall FNR. Table II confirms this effect: averaged over five flood events (excluding Bremen, which contains no FB/FV), the average OOD abstention power is 63.3%, peaking at 92.0% for the Beledweyne flood and dipping to 33.7% for Caprivi. The average OOD FDR is 2.6%, ranging from 0.9% (Itaqui) to 6.9% (Iwaki). As a result, the average water detection FNR drops from 36.9% to 28.5% after abstention is applied. Visual inspection of Fig. 8 shows that abstention occurs primarily

TABLE III
EVALUATION OF ID PREDICTION PERFORMANCE ACROSS VARYING α VALUES WITH FIXED $\delta = 0.1$

Study Site	Risk Tolerance Level	Marginal miscoverage	Water miscoverage	Non-water miscoverage
Bremen, Germany	Singleton prediction	3.3%	11.4%	2.2%
	$\alpha = 0.1$	2.4%	8.6%	1.5%
	$\alpha = 0.05$	1.0%	3.8%	0.6%
	$\alpha = 0.01$	0.0%	0.3%	0.0%
Caprivi, Namibia	Singleton prediction	7.6%	11.5%	5.0%
	$\alpha = 0.1$	6.0%	9.5%	3.6%
	$\alpha = 0.05$	2.9%	5.1%	1.5%
	$\alpha = 0.01$	0.3%	0.6%	0.1%
Beledweyne, Somalia	Singleton prediction	5.5%	8.2%	5.2%
	$\alpha = 0.1$	4.2%	6.1%	4.2%
	$\alpha = 0.05$	2.1%	2.3%	2.1%
	$\alpha = 0.01$	0.4%	0.1%	0.5%
Itaqui, Brazil	Singleton prediction	1.6%	3.3%	1.1%
	$\alpha = 0.1$	1.1%	2.2%	0.8%
	$\alpha = 0.05$	0.3%	0.6%	0.3%
	$\alpha = 0.01$	0.0%	0.0%	0.0%
Houston, US	Singleton prediction	6.3%	47.5%	1.8%
	$\alpha = 0.1$	5.1%	39.6%	1.3%
	$\alpha = 0.05$	2.7%	20.8%	0.7%
	$\alpha = 0.01$	0.3%	2.1%	0.1%
Iwaki, Japan	Singleton prediction	2.0%	34.9%	1.2%
	$\alpha = 0.1$	1.4%	28.4%	0.8%
	$\alpha = 0.05$	0.6%	13.1%	0.3%
	$\alpha = 0.01$	0.1%	1.4%	0.0%
Average	Singleton prediction	4.4%	19.5%	2.7%
	$\alpha = 0.1$	3.4%	15.7%	2.0%
	$\alpha = 0.05$	1.6%	7.6%	0.9%
	$\alpha = 0.01$	0.2%	0.8%	0.1%

over FB and FV regions, as intended, along with some pixels along water boundaries. Additional abstention is observed in complex urban areas and mountainous terrain affected by SAR-specific characteristics (e.g., foreshortening in the Iwaki case). These abstentions are not detrimental, as these areas are inherently difficult for the model to predict reliably. Some FB/FV regions, however, are not identified as OOD because their latent features resemble those of seen normal urban or vegetated classes, resulting in high feature density and low EU. This observation aligns with previous studies [10], [68], which indicate that single-channel SAR intensity alone is often insufficient to distinguish FB and FV from other land covers. The extent of abstention also varies with the risk control level, as further explored in Section VI-D.

Table III reports the marginal miscoverage (including both open water and nonwater classes), along with class-wise miscoverage under standard singleton predictions and conformal risk control prediction sets. With conformal risk control, the average marginal miscoverage decreases from 4.4% to 1.6%, average water miscoverage declines from 19.5% to 7.6%, and average nonwater miscoverage from 2.7% to 0.9%. A comparison of Figs. 4 and 8 shows that pixels with high AU are generally assigned a two-element prediction set {Water, Nonwater}, rather than a confident singleton label. Since prediction under conformal risk control is governed by a single parameter $\hat{\lambda}$, the corresponding threshold in softmax entropy can be used to map uncertainty to prediction set size. Predictions with entropy above this threshold are converted into ambiguous sets, thus establishing an interpretable link between uncertainty and prediction confidence. This mechanism is particularly useful in ambiguous regions. For instance, open floodwater may not always exhibit characteristically low backscattering, as observed in agricultural areas (Bremen), wind-roughened waters (Itaqui), and cluttered urban scenes (Houston, Iwaki). In such situations, the model assigns the set {Water, Nonwater}, preserving predictive coverage. Conversely, land surfaces with dark SAR tones—such as drylands in Somalia or paved urban surfaces—are often misclassified as water in standard prediction, but are now handled more cautiously. Nevertheless, as shown in Table III, water miscoverage for the Houston and Iwaki events remains above the nominal risk level $\alpha = 0.05$. This can be attributed to several factors: 1) calibration data primarily represent rural open flood events, underrepresenting complex urban environments; 2) a single $\hat{\lambda}$ was used for marginal risk control, rather than class-conditional calibration; and 3) even under class-conditional calibration, test-conditional validity (e.g., controlling miscoverage for a specific test set) is not guaranteed—the framework only assures training- and calibration-conditional validity. These limitations are further discussed in Section VI-D.

Overall, the results demonstrate that conformal risk control enhances both prediction robustness and interpretability. It yields three types of outputs with statistical guarantees: 1) **single-element sets**, e.g., {Water} or {Nonwater}, which indicate high-confidence predictions; 2) **two-element sets**, {Water, Nonwater}, which identify ambiguous cases with high AU; and 3) **empty sets**, $\emptyset$, which flag OOD regions where no reliable prediction can be made. These three

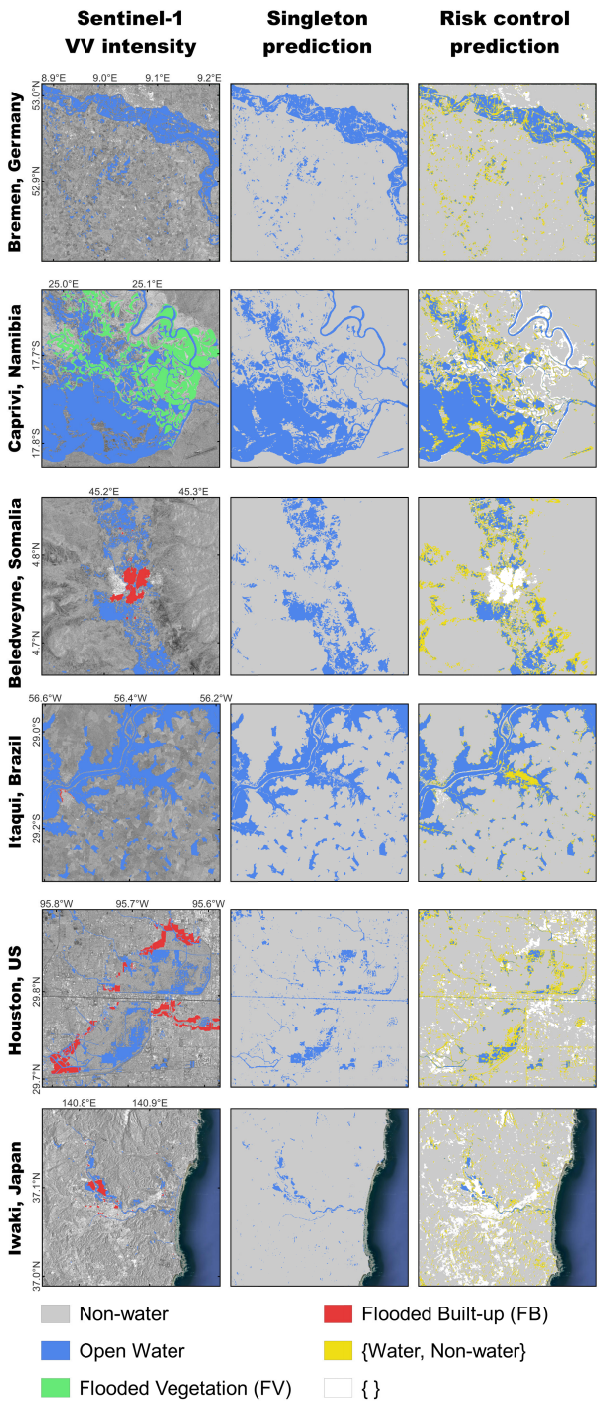

Fig. 8. Comparison between standard singleton predictions and prediction sets generated via conformal risk control.

prediction-set outcomes translate directly into risk-calibrated, actionable guidance for both decision-makers and model developers. Single-element prediction sets provide high-confidence predictions for operational tasks such as delineating evacuation zones, prioritizing areas for field deployment, and supporting rapid response during an unfolding event. Ambiguous predictions (two-element sets) highlight locations where additional information dimensions should be consulted before action. Although the model's underlying singleton prediction may still be correct in these regions, safer decisions require auxiliary evidence, for example, integrating outcomes from other models, requesting drone-based visual confirmation, or inspecting high-resolution DEMs. For model developers, these regions reveal opportunities to reduce prediction ambiguity by enriching the input feature space (e.g., dual-polarized SAR, multispectral imagery, DEMs, land-cover layers). Empty-set outputs are especially valuable for risk-averse operations. They explicitly flag regions where the model cannot provide a reliable prediction, indicating that decisions should rely solely on external sources such as ground reports or outcomes from complementary models. From a development standpoint, empty-set regions point to structural limitations of the current model or training distribution, motivating task reformulation (e.g., moving from binary to multiclass segmentation) and the collection of targeted training data to improve the representation of such cases. Prediction sets can also enhance the use of the model's singleton predictions. For example, in weighted ensembling, singleton predictions within single-element sets can be assigned high weight; those overlapping two-element sets may be given moderate weights; and predictions associated with empty sets may receive minimal or zero weight. In practice, this triaged structure enables emergency managers to clearly differentiate where automated outputs are trustworthy, where caution and auxiliary evidence are warranted, and where human judgment or alternative models must take precedence.

VI. Discussion

A. Revisit BNN for Uncertainty Quantification

In this section, we revisit the widely used BNN framework (e.g., MC Dropout and DEs) to understand why they fail to capture high EU in the presence of FB and FV in the context of open (flood) water detection. Both MC Dropout and DEs estimate EU via the MI between model outputs and parameters across multiple forward passes. This measure is derived from the softmax output $p(y|x,\theta) = p(x|y,\theta)p(y)/p(x)$ and describes parameter uncertainty through variation in predictions. However, this approach struggles under distributional shifts (i.e., OOD samples) or model misspecification. The core limitation lies in relying solely on criteria based on $p(y|x)$ [see (1)]. Because normalizing by $p(x)$ implicitly ignores the probability of encountering a given input x, OOD samples with very low $p(x)$ can receive high-confidence predictions $p(y|x)$ for the most likely label [69]. In our case, multiple forward passes with MC Dropout and DEs consistently produce overconfident predictions on FB and FV samples—misclassifying them as nonwater. Although these samples lie far from the training distribution [e.g., have low $p(x)$], their MI values remain low, failing to reflect appropriate EU. Prior work provides theoretical and empirical support for this phenomenon. The study by [70] argues that uncertainty estimation should focus on $p(x|y)$, rather than $p(y|x)$, to enable genuinely cautious predictions. Similarly, [71] advocates for incorporating density uncertainty layers to improve EU quantification on OOD data. In line with these perspectives, [41] emphasizes that the EU must account for model misspecification and that data or feature density is essential for effective and meaningful

TABLE IV
ABLATION STUDY OF UNCERTAINTY QUANTIFICATION IN THE PROPOSED DENSITY-AWARE NEURAL NETWORK

Architecture	Ablation		ECE (↓)	AUROC (↑)
	Spectral Normalization	PCA		
UNet34	✗	✗	0.014	0.901
	✓	✗	0.014	0.911
	✓	✓	0.014	0.921

TABLE V
COMPARISON OF TRAINING AND INFERENCE TIMES FOR DIFFERENT MODELS

Model	Training time for 200 epochs (min)	Inference time on a 4096×4096-pixel image (s)
MC Dropout	9	14
Deep Ensembles	90	7
Density-aware Neural Network	12	6

estimation. Our experimental results support these findings. In the flood mapping context, feature density plays a critical role in detecting OOD inputs and enabling accurate uncertainty quantification and abstention. While it may be argued that model misspecification is less problematic in DL—due to the expressive hypothesis space of neural networks—this space remains constrained not only by the network architecture but also through training configurations, loss functions, and validation datasets [41]. In complex flood mapping scenarios (e.g., urban or vegetated environments), the conditional probability $p(x|y)$ exhibits bimodal behavior across open water and FB/FV classes. This indicates that a binary segmentation model (e.g., distinguishing only between open water and nonwater) is inherently misspecified for such tasks. A multiclass formulation (e.g., separating open water, nonwater, and FB/FV) is therefore more appropriate. While a broad range of EU methods has been proposed in the ML literature, exhaustive benchmarking is not the primary objective of this study. We focus on widely adopted approaches that estimate EU using entropy-based measures, such as MI. Our results and discussion highlight the limitations of these entropy-based measures under realistic distributional shifts encountered in SAR-based flood mapping. These findings also underscore the need for continued investigation of density-based methods to capture EU arising from model misspecification more effectively.

B. Ablation Experiments of Density-Aware Neural Network

As described in Section III-B, two key components were added to the UNet34 architecture to facilitate EU quantification: 1) spectral normalization applied to each convolutional layer and 2) PCA applied to the learned feature density $q(z)$. Table IV summarizes the ablation results, reported using the average AUROC for OOD detection across five test flood events. The results indicate that incorporating spectral normalization increases the average AUROC from 0.901 to 0.911, and adding PCA yields an additional improvement to 0.921. Interestingly, even the baseline UNet34 (without spectral normalization) already preserves meaningful distances in the learned feature space. One plausible explanation is that water and nonwater samples in the training set are relatively easy to separate, which reduces the need for highly abstract feature transformations. Moreover, the mono-modal input (i.e., single-channel VV intensity) may allow the model to implicitly retain data-space distances within the latent representation, even in the absence of an explicit Lipschitz constraint. This observation aligns with recent findings that feature density in the latent space can serve as an effective proxy for EU, even without applying strong smoothness regularization, such as gradient penalties or spectral normalization [72]. While spectral normalization and other smoothness-enforcing mechanisms can empirically improve robustness in preserving input distance, the theoretical interactions among model architecture, loss function, and optimization dynamics remain insufficiently understood and warrant further investigation [73].

C. Computational Efficiency and Complexity Analysis

Table V presents the computation time of the evaluated models. During the training phase, the density-aware neural network incurs a modest overhead due to the fitting of PCA and GDA. DEs are the most computationally demanding because multiple independent models must be trained (although the training time can be reduced through distributed training). For inference, the density-aware neural network is the most efficient because it requires only a single forward pass, and the subsequent PCA and GDA transformations add negligible latency. In contrast, both MC Dropout and DEs require multiple forward passes, resulting in higher inference costs. The conformal risk control procedures for OOD abstention and ID prediction sets generation are model-agnostic and depend only on precomputed uncertainties and the calibration set. In our experiments, OOD abstention on a 4096×4096-pixel input required 5 min 20 s during inference; this step can be accelerated through tile-based parallelization for large-scale operational deployment. The asymptotic adjustment of marginal p-values introduces an additional computational complexity of $O(m(\log\log n)^{1/2}/\sqrt{n})$, where m is the number of test samples and n is the number of calibration samples. For ID prediction sets, the most computationally intensive step is the search for the optimal parameter $\hat{\lambda}$ during calibration (18 min in our experiment). After calibration, inference incurs only a lightweight overhead of $O(mc)$ on top of the standard singleton prediction, where m is the number of test samples and c is the number of classes.

D. Sensitivity in Conformal Risk Control

To assess the impact of risk tolerance settings on prediction performance, we conducted a sensitivity analysis by varying the predefined marginal risk level $\alpha \in \{0.01, 0.05, 0.1\}$, while holding $\delta = 0.1$ fixed. This configuration reflects practical deployment scenarios where users often aim to enforce a desired risk tolerance with high probability. The calibration-conditional conformal p-value is constructed to control the false discovery risk associated with abstaining on OOD regions. Table II quantifies the abstention performance using the same metrics introduced in Section V-B. As shown in Fig. 9, increasing the risk tolerance α improves the abstention power, enabling broader exclusion of uncertain OOD

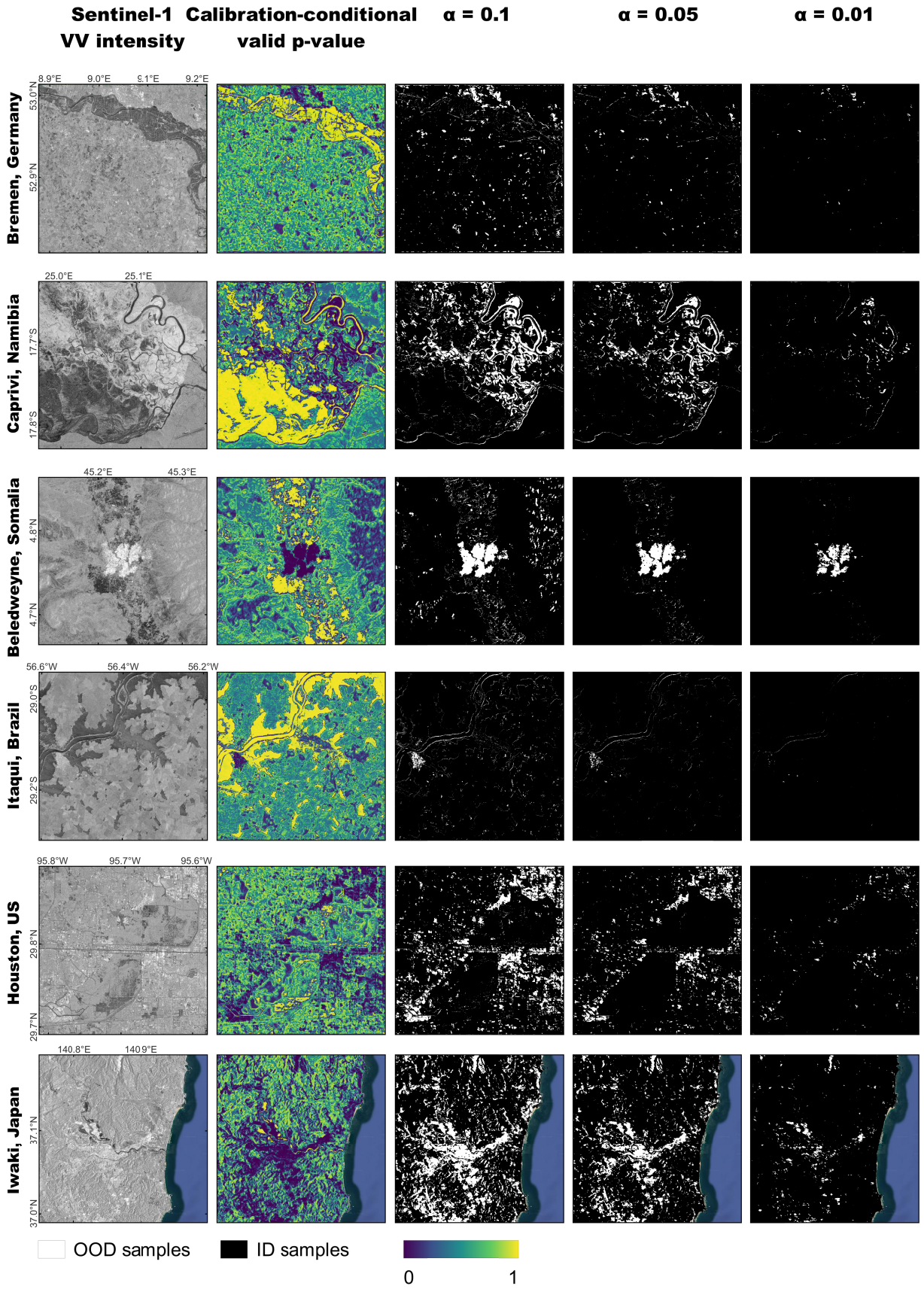

Fig. 9. OOD abstention results under varying risk tolerance levels α. Smaller α values yield more conservative OOD masks.

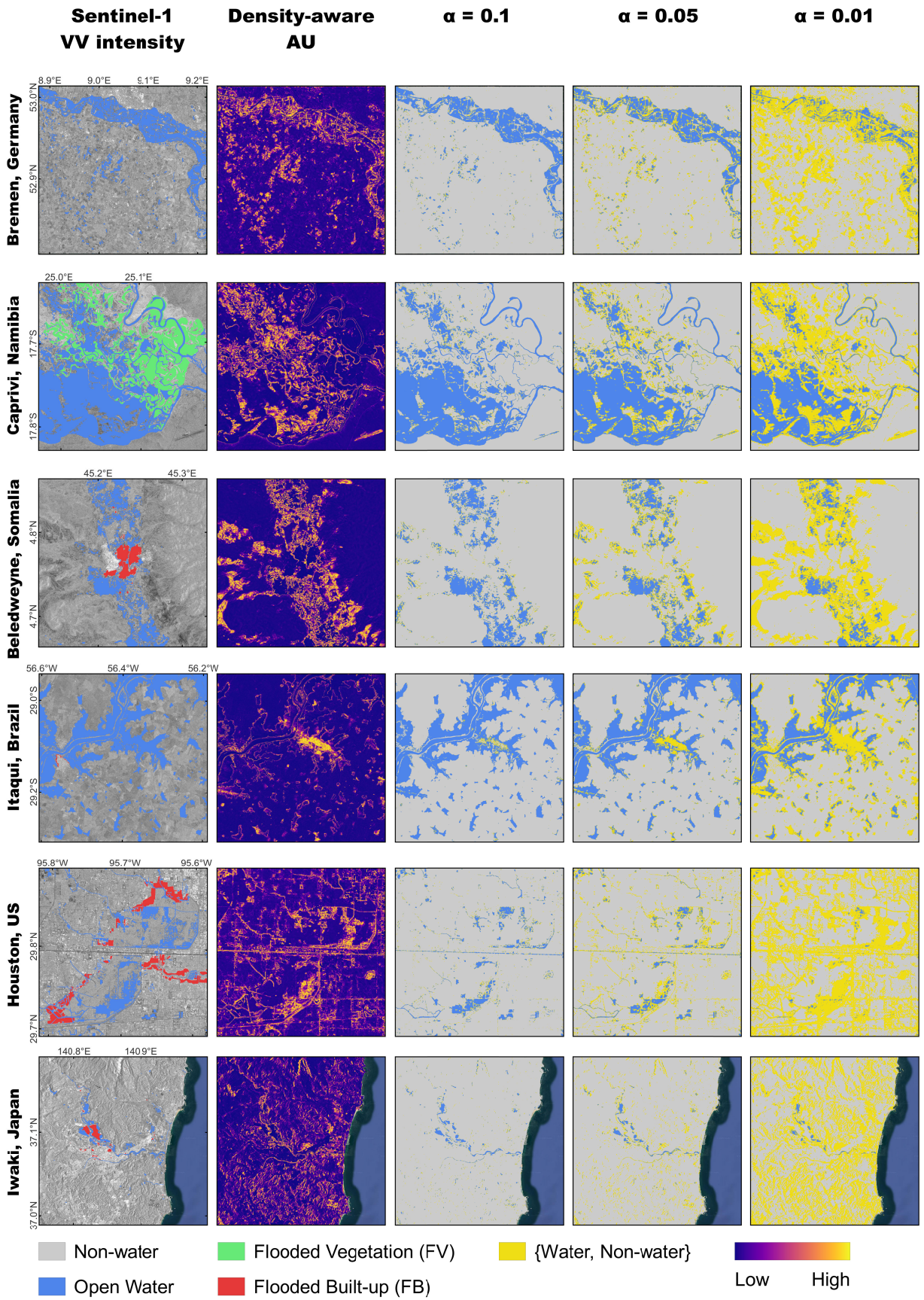

Fig. 10. Prediction sets under varying risk tolerance levels α. Smaller α values result in a larger proportion of two-element (ambiguous) prediction sets to satisfy the coverage guarantee.

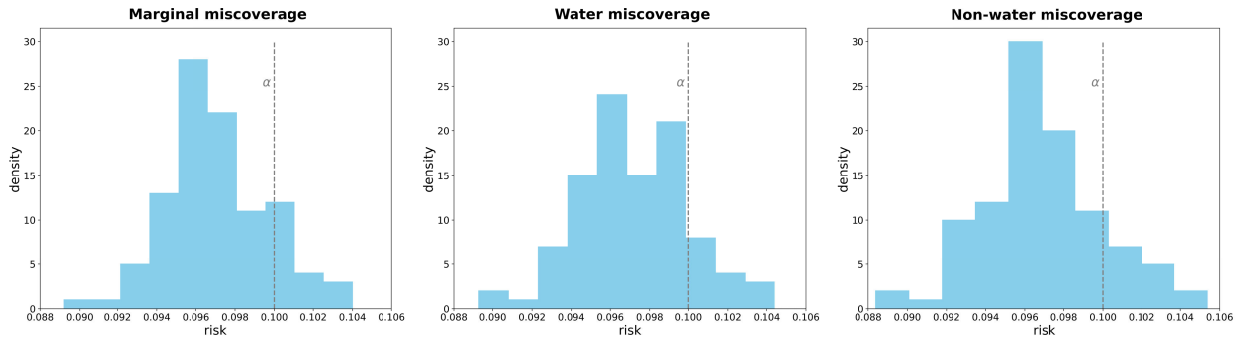

Fig. 11. Histograms of miscoverage rates over 100 random splits of flood image patches under a risk level ($\alpha = 0.1$ and $\delta = 0.1$).

samples. However, this gain comes at the expense of higher false discovery risk—resulting in the unintended abstention of samples from ID regions, particularly open water areas where the model might otherwise have made correct predictions. This tradeoff is further reflected in water detection performance: as α increases, the FNR in challenging FB and FV regions decreases, but the corresponding increase in OOD FDR indicates a larger proportion of true floodwater is unnecessarily excluded. Therefore, selecting an appropriate α involves balancing informativeness and caution—retaining confident predictions while abstaining from highly epistemically uncertain regions that are prone to errors. In complex urban settings such as the Houston flood, a higher α (e.g., 0.1) may be preferred to more aggressively filter unreliable predictions, despite the potential loss of relevant flood information. AU and conformal prediction sets at different risk levels are visualized in Fig. 10, and Table III reports the corresponding miscoverage rates. These results confirm that the conformal risk control framework effectively regulates both marginal and class-conditional miscoverage within user-defined risk levels, conditioned on the calibration set.

As previously discussed, we emphasize that conformal risk control guarantees calibration-conditional validity rather than stronger test-conditional validity. To investigate the robustness of risk control across test distributions, we performed 100 random splits of flood image patches, using 30 000 pixels for calibration and 20 000 for evaluation in each split. As shown in Fig. 11, the framework effectively maintains the target risk level. However, in some challenging test cases—such as the Houston and Iwaki floods—water miscoverage rates exceed the nominal risk level. This violation is likely attributable to a domain shift, where the calibration set (primarily composed of rural scenes) fails to adequately represent the characteristics of complex urban test environments. In real-world applications, the selection of α should take into account the environmental context. For instance, in arid or semi-arid regions where floodwater is often overdetected, or in complex urban areas where underdetection is more prevalent due to ambiguous open water signatures, a lower α (e.g., 0.01) may be preferred to reduce miscoverage—whether for nonwater or water classes, respectively. As illustrated in Fig. 10, decreasing α increases the prevalence of two-element prediction sets (i.e., Water, Nonwater), thereby reducing both types of miscoverage. In the extreme case where $\alpha = 0$, all predictions would yield the two-element set, reflecting fully ambiguous and making the output uninformative. Therefore, excessively low α values should be avoided to ensure the model outputs remain informative and operationally valuable.

E. Limitations and Future Work

A key limitation of density-based uncertainty quantification lies in the incomplete understanding of the effects of smoothness regularization in neural networks. While the bi-Lipschitz constraint—imposed through a combination of residual connections and spectral normalization—has been theoretically and empirically validated [43], [48], its interaction with other aspects of model design remains insufficiently studied. These aspects include alternative network architectures beyond CNNs (e.g., transformers), different loss functions, and various optimization processes [73]. Furthermore, it remains an open question whether new definitions and metrics are needed to more effectively characterize smoothness in high-capacity models. Addressing these questions necessitates deeper investigation, both empirically and theoretically.

Another limitation concerns the conformal risk control framework's calibration-conditional validity, which does not ensure *pointwise test-conditional coverage*—that is, coverage conditioned on a specific test input $X_{\text{test}} = x$. This stronger form of guarantee is often more desirable in operational settings where prediction reliability for individual samples is critical. Although such guarantees are theoretically achievable in discrete domains (by calibrating separately for each discrete value of X), they are provably unachievable in continuous settings without strong distributional assumptions [29]. Nevertheless, promising relaxations of pointwise test-conditional coverage have been proposed to bridge this gap. For example, subgroup-wise test-conditional coverage can be attained by partitioning the continuous input space into bins or clusters and ensuring coverage within each subgroup [74], [75]. Further investigation into these relaxed notions—alongside methods such as incorporating local weights [76], [77] or reformulating the framework from a Bayesian perspective [78], [79]—represents a promising direction for enhancing practical utility.

Beyond these theoretical limitations, several practical extensions are essential for broadening the applicability of the proposed framework in remote sensing applications. Although

the present study addresses a binary prediction task, a natural direction for future work is to extend the framework to multiclass segmentation problems. Another important avenue for expansion is its adaptation to regression tasks, such as the retrieval of biophysical variables. Since regression involves a continuous target space, the latent feature density estimation used in the current density-aware neural network is not directly applicable. Alternative approaches, such as normalizing flows or other continuous density models, will be required [80], [81], together with task-appropriate risk functions for generating ID prediction intervals. In addition, the integration of multimodal and multisensor data has become increasingly prevalent in remote sensing. Recent advances in cross-attention mechanisms and transformer-based fusion architectures have demonstrated strong performance across a range of tasks [82], [83], [84]. How to impose and preserve bi-Lipschitz constraints in such multistream fusion networks remains an open research question. Finally, while the conformal risk control procedures employed in this work are computationally tractable, the calibration steps introduce non-negligible overhead. Developing more efficient and incremental calibration strategies will be important to support large-scale or near-real-time operational deployments.

VII. Conclusion

This study proposed a unified framework for uncertainty quantification and communication in SAR-based flood mapping. The framework comprises two principal components: a density-aware neural network for uncertainty estimation, and a conformal risk control mechanism for uncertainty communication. The first component utilizes an image segmentation model (UNet34) augmented with spectral normalization to enforce a bi-Lipschitz constraint and preserve data distances in the input space. EU is estimated via feature density in the latent space, reflecting the model's confidence under distributional shift, while AU is quantified using softmax entropy, capturing intrinsic data ambiguity. The second component, conformal risk control, enables the construction of statistically valid prediction sets under a user-specified risk level (α, δ). Samples exhibiting high EU (i.e., OOD) are abstained from prediction, whereas ID samples receive either single-element prediction sets (i.e., {Water} or {Nonwater}) or two-element prediction sets (i.e., {Water, Nonwater}) based on their AU. This formulation guarantees that the miscoverage risk does not exceed α with probability at least $1 - \delta$, under calibration-conditional validity.

Empirical evaluations across diverse flood scenarios—ranging from rural floodplains to urban environments and vegetated areas—demonstrated the framework's effectiveness in mitigating false negatives by abstaining from regions with elevated EU, such as FB and vegetated areas. For ID samples, the resulting prediction sets improved reliability and interpretability relative to standard singleton predictions, thereby supporting more informed downstream decision-making. Despite its empirical strengths, the framework presents several avenues for future research. The theoretical interplay among smoothness constraints, network architectures, loss functions, and optimization processes remains insufficiently understood and thus warrants deeper investigation. In addition, while the current conformal risk control ensures calibration-conditional validity, extensions toward pointwise or subgroup-wise test-conditional validity may provide stronger guarantees in heterogeneous environments. Finally, although this work focuses on SAR-based flood mapping, the proposed methodology is broadly applicable to other remote sensing tasks involving image classification or segmentation, and can be adapted for continuous-domain problems such as regression.

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